

Customer Lifetime Value (CLV) Prediction

Using Regression Analysis

A Statistical Modeling Study on E-Commerce Transactional Data

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Dataset	~500,000 transaction records; ~3,000–5,000 customer-level observations
Target Variable	CLV — total revenue in the next 90 days
Primary Model	Multiple Linear Regression (OLS)
Comparison Model	Random Forest Regressor (200 estimators)
Key Metric	Root Mean Squared Error (RMSE)
Key Finding	Top 20% of customers contribute 78% of predicted revenue

1. Dataset Selection and Problem Statement

The dataset used in this study is an e-commerce transactional dataset containing approximately 500,000 transaction-level records. Each record includes CustomerID, InvoiceDate, Quantity, UnitPrice, and product details. The dataset was preprocessed by removing null CustomerIDs, filtering out negative quantities and unit prices, and converting invoice dates to datetime format. A revenue column was derived as the product of Quantity and UnitPrice.

For regression analysis, the dataset was aggregated at the customer level, yielding approximately 3,000–5,000 observations. A time-based train/test split was applied at the 75th percentile of InvoiceDate to preserve temporal ordering. The dependent variable (CLV) was defined as the total revenue a customer generates in the 90 days following the training period's end date.

Independent Variables (Feature Set)

- Recency — Days since the customer's last purchase
- Frequency — Number of unique invoices
- Monetary Total — Total revenue generated
- Average Order Value (AOV) — Mean revenue per transaction
- Product Diversity — Number of unique products purchased
- Tenure — Days between first and last purchase
- Purchase Velocity — Frequency divided by (Tenure + 1)

The objective was to apply regression techniques to predict CLV and quantify the relationship between customer behavioral features and future revenue.

2. Model Specification

A multiple linear regression model was specified as the primary analytical tool:

$$CLV = \beta_0 + \beta_1(\text{Recency}) + \beta_2(\text{Frequency}) + \beta_3(\text{Monetary}) + \beta_4(\text{AOV}) + \beta_5(\text{Diversity}) + \beta_6(\text{Tenure}) + \beta_7(\text{Velocity}) + \varepsilon$$

Linear regression was selected because it provides interpretable coefficients for each feature, supports statistical hypothesis testing (t-tests and F-tests), and establishes a transparent baseline against which more complex models can be benchmarked.

Regression Assumptions

- Linearity between each predictor and the CLV response
- Independence of error terms across observations
- Homoscedasticity — constant variance of residuals
- Normality of residuals for valid inference

3. Parameter Estimation

Coefficients were estimated using Ordinary Least Squares (OLS) via the statsmodels library, with a constant term added to the design matrix. The table below summarizes the estimated coefficients, standard errors, and statistical significance for each predictor.

Variable	Coefficient	p-value	Significance
Constant	140.29	0.412	Not Significant
Recency	-0.0735	0.922	Not Significant
Frequency	32.4776	0.024	Significant *
Monetary Total	0.1766	0.000	Significant ***
Avg Order Value	-0.7487	0.114	Not Significant
Product Diversity	2.1988	0.047	Significant *
Tenure	-1.7025	0.047	Significant *
Purchase Velocity	-23.8406	0.860	Not Significant

The most impactful predictor is Monetary Total ($\beta = 0.1766$, $p < 0.001$), indicating that every additional unit of historical spending is associated with a meaningful increase in predicted CLV. Frequency also shows a strong positive effect ($\beta = 32.48$, $p = 0.024$), reflecting that more frequent buyers tend to generate greater future value. Product Diversity contributes positively ($\beta = 2.20$, p

= 0.047), suggesting that breadth of purchasing behavior is a signal of engagement. Tenure carries a small but statistically significant negative coefficient ($\beta = -1.70$, $p = 0.047$). Recency, AOV, and Purchase Velocity were not found to be statistically significant in this model.

4. Model Significance

Overall Model Fit

The F-statistic is 86.88 with a p-value of 9.52e-116, conclusively establishing that the regression model as a whole is statistically significant. The model explains 16.5% of the variance in CLV ($R^2 = 0.165$, Adjusted $R^2 = 0.163$). While this level of explanatory power is modest, it is common in behavioral prediction problems with highly skewed outcome distributions.

R^2	Adj. R^2	F-Statistic	Prob (F)
0.165	0.163	86.88	9.52e-116

5. Residual Analysis and Diagnostics

A thorough residual analysis was conducted to evaluate whether the four classical regression assumptions hold for this model.

Residual Plot

The scatter plot of residuals against fitted values reveals non-random patterns and the presence of extreme outlier values. The spread of residuals increases substantially at higher predicted values, indicating a violation of constant variance. A small number of observations contribute disproportionately large residuals, consistent with the highly skewed CLV distribution in the raw data.

Normality of Residuals

The residual histogram exhibits an extremely narrow, leptokurtic distribution centred near zero but with very long tails extending to $\pm 40,000$. The Q-Q plot confirms this: observations in the tails depart dramatically from the theoretical normal line. Formal statistics reinforce the visual evidence — the Omnibus test statistic is 9522.015 ($p \approx 0$), the skewness is 46.07, and the kurtosis is 2368.09, all far beyond normal-distribution benchmarks. The Jarque-Bera statistic exceeds 700 million, confirming severe non-normality.

Homoscedasticity

The Breusch-Pagan test yields a p-value of approximately 2.08e-156, providing overwhelming evidence against constant variance. Heteroscedasticity of this magnitude is expected when the response variable is heavily right-skewed, as is the case with CLV distributions in e-commerce data.

Independence of Errors

The Durbin-Watson statistic is 2.001, very close to the ideal value of 2.0. This indicates no meaningful autocorrelation among residuals, so the independence assumption is satisfactorily met.

Multicollinearity

The condition number is reported as 2.51×10^4 , which is moderately large and may indicate the presence of multicollinearity or numerical ill-conditioning among some predictors. This should be investigated further if the model is deployed in a production setting.

Diagnostic Summary

Assumption	Test / Evidence	Result
Linearity	Residual plot	Violated — non-random patterns
Normality	Histogram, Q-Q plot, JB Test	Violated — severe skew & kurtosis
Homoscedasticity	Breusch-Pagan ($p \approx 2.08e-156$)	Violated — heteroscedasticity confirmed
Independence	Durbin-Watson = 2.001	Satisfied

6. Model Improvement — Random Forest

Given that three of four regression assumptions were violated, a Random Forest Regressor was implemented as a non-linear alternative. The model was trained with 200 decision tree estimators using all 7 features on the same training set, with `n_jobs=-1` for parallel processing.

Model	RMSE
Linear Regression	1943.14
Random Forest	1729.40 ✓ Better

The Random Forest model reduces RMSE from 1943.14 to 1729.40 — an improvement of approximately 11%. This confirms that non-linear feature interactions and complex relationships between customer behavior variables contribute meaningfully to predictive accuracy. Importantly, Random Forest is not constrained by the distributional assumptions of OLS, making it more robust to the skewed CLV distribution and heteroscedastic errors observed in this dataset.

After refitting on the full dataset, customers were segmented into three tiers using quantile-based binning of predicted CLV: Low, Medium, and High. Analysis of the top 20% of customers by predicted CLV revealed that this segment accounts for 78.00% of total predicted revenue — a Pareto-consistent pattern common in retail and e-commerce.

7. Conclusion

This study applied regression analysis to predict customer lifetime value from behavioral features derived from e-commerce transaction history. The linear regression model, while statistically

significant at the overall level, explained only 16.5% of CLV variance and violated three of four OLS assumptions — most critically normality and homoscedasticity.

Despite these limitations, the model provides actionable directional insights: monetary spending history and purchase frequency are the strongest drivers of future customer value, while recency and average order value were not significant predictors in this context. The negative tenure coefficient, though counterintuitive, may reflect the lower marginal return of older customers who have already been captured by the spending and frequency variables.

The Random Forest model demonstrates that non-linear ensemble methods better capture the complexity of customer behavior data. The 78% revenue concentration among the top 20% of customers provides a direct strategic insight: targeted retention and upsell investments in this segment would yield disproportionate returns.

8. Limitations

- The dataset does not include session-level or behavioral clickstream data, which could substantially improve predictive accuracy
- The CLV distribution is highly right-skewed; a log or Box-Cox transformation could mitigate violations of normality and homoscedasticity
- Classical regression assumptions are not fully satisfied, limiting the validity of confidence intervals and p-values under OLS
- External factors such as promotional campaigns, seasonality, and product category trends were not incorporated
- The 90-day prediction window was fixed; sensitivity to this design choice was not assessed
- Potential multicollinearity among RFM features (Recency, Frequency, Monetary) was not formally addressed through dimensionality reduction

This report was prepared based on an e-commerce transactional dataset. All analysis was performed in Python using statsmodels, scikit-learn, and pandas.